Dawei Sun, M.D., Ph.D.

Anesthesiologist & Clinical Researcher

dawei.sun.md@gmail.com | daweisun.com

EDUCATION AND TRAINING

M.D. - Anesthesiology
Wenzhou Medical University
October 2010 - July 2015

Ph.D. - Anesthesiology
Shanghai Jiao Tong University
September 2015 - June 2020

Visiting Scholar - Neurology and Pediatrics
University of California San Francisco, United States
October 2018 - October 2019

Residency - Anesthesiology
The Second Affiliated Hospital, Zhejiang University School of Medicine
September 2020 - August 2023
Training included an intensive care unit (ICU) rotation.

Clinical Postdoctoral Fellow
Zhejiang University
September 2020 - February 2024

LICENSURE AND PROFESSIONAL QUALIFICATIONS

ECFMG Certification - Educational Commission for Foreign Medical Graduates, United States, 2026

USMLE Step 3 - Scheduled

Associate Senior Professional Title in Research (Associate Researcher), Zhejiang Province, China, 2025

Intermediate Professional Qualification in Anesthesiology, China, 2024

Physician Qualification Certificate and Physician Practice Certificate, China, 2016

HONORS AND AWARDS

One of 10 abstracts selected for the Best of Abstracts: Basic Science oral session from more than 1,000 submissions, American Society of Anesthesiologists Annual Meeting, 2019.

PATENT

Sun D, Luo G, Yao Y, Yan M. Continuous oxygen saturation monitoring probe for insertion into a central venous catheter. Chinese Utility Model Patent, ZL 2024 2 3305545.1. Granted April 21, 2026.

A probe designed for placement within a central venous catheter to continuously monitor central venous oxygen saturation (ScvO2), intended to support real-time assessment of the balance between oxygen delivery and consumption.

PROFESSIONAL EXPERIENCE

Associate Researcher
Sub-Laboratory of the State Key Laboratory of Membrane Biology, Hangzhou Lin'an District Qingshan Lake Future Industry and Technology Innovation Center, Hangzhou, China
November 2024 - Present
Research on cerebral ischemia and neuroprotection, with a focus on autophagy and mitophagy.

Anesthesiologist
The Second Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China
September 2023 - October 2024

- Provided perioperative anesthesia care across a broad range of surgical procedures, with senior consultation for complex or high-risk cases.

- Staffed the preoperative anesthesia clinic: evaluation, risk assessment, patient optimization, and anesthetic planning.

- Helped establish the preoperative anesthesia clinic, including patient screening criteria and assessment workflows.

- Supported clinic IT system optimization, critical-value alerts and escalation processes, and preoperative communication across departments.

- Managed induction, intraoperative monitoring and management, emergence, and postoperative handoff.

- Performed ultrasound-guided regional anesthesia and vascular access, including major peripheral nerve blocks, arterial catheterization, and central venous access.

- Participated in operating-room orientation and case-based teaching for international medical students at Zhejiang University.

U.S. CLINICAL OBSERVERSHIP

Clinical Observer
University of Utah Hospital, United States
November 2026 - Four weeks
Department of Anesthesiology, Perioperative & Pain Medicine

Clinical Observer
Torrance Memorial Hospital, Torrance, California, United States
October 2026 - Four weeks
Department of Anesthesiology

RESEARCH FUNDING

Principal Investigator - NSFC Young Scientists Fund
National Natural Science Foundation of China
2024 - 2026
Neuroprotection and mechanism of HIF-2α on immature brain hypoxic-ischemic injury by regulating mitophagy.

PEER-REVIEWED PUBLICATIONS

Ordered by CV publication year, including online-first dates. First, co-first, and corresponding authorship are marked.

1. Wei Y, **Sun D**, Ng S, He Y, Liu X, Wang J. Cardiovascular health and life expectancy free of atherosclerotic cardiovascular disease among patients with type 2 diabetes: a cohort study. European Journal of Preventive Cardiology. 2025;:zwaf742. DOI: 10.1093/eurjpc/zwaf742.

2. Wang B, Li S, Wang C, Zhu J, Jin Y, **Sun D**, Lang J. Effect of sevoflurane post-conditioning on postoperative cognitive function in carotid endarterectomy patients: a randomized, double-blind, controlled trial. Frontiers in Medicine. 2025;12:1716370. **(Corresponding author)** DOI: 10.3389/fmed.2025.1716370.

3. Liu L, Li F, Wei Y, Luo L, Shen L, Li J, Sun N, Qian B, **Sun D**. Efficacy and Safety of Ciprofol Alone versus Ciprofol with Fentanyl for Upper Gastrointestinal Endoscopy: A Randomized, Double-Blind, Controlled Trial. Drug Design, Development and Therapy. 2025;19:5231-5241. **(Corresponding author)** DOI: 10.2147/dddt.s516064.

4. Chen Y, Yao Y, **Sun D**, Wang Y, Wu C, Zhao L, Tang Z, Zhang F. Advances in Monitoring and Reversal Strategies for the Novel Anticoagulants Factor XI Inhibitors. Chinese General Practice. 2025;28:379-384.

5. Wei Y, **Sun D**, Jaiswal S, He Y, Liu X, Wang J. Association of lifestyle with valvular heart disease progression and life expectancy among elderly people from different socioeconomic backgrounds. BMC Medicine. 2024;22(1):367. DOI: 10.1186/s12916-024-03576-9.

6. Wang R, Xiao J, Gao Q, Xu G, Ni T, Zou J, Wang T, Luo G, Cheng Z, Wang Y, Tao X, **Sun D**, Yao Y, Yan M. Predictive modeling for identifying infection risk following spinal surgery: Optimizing patient management. Experimental and Therapeutic Medicine. 2024;28(1):281. DOI: 10.3892/etm.2024.12569.

7. Tao X, Luo G, Xiao J, Yao Y, Gao Q, Zou J, Wang T, Cheng Z, **Sun D**, Yan M. Chronic Postsurgical Pain Following Lung Transplantation: Characteristics, Risk Factors, Treatment, and Prevention: A Narrative Review. Pain and Therapy. 2024;13(4):719-731. DOI: 10.1007/s40122-024-00615-4.

8. Ni T, Yao Y, Zhou X, Lv T, Zou J, Luo G, Yang J, **Sun D**, Gao Q, Wang T, Wang R, Tao X, Yan M. Postinduction Hypotension and Adverse Outcomes in Older Adults Undergoing Transcatheter Aortic Valve Replacement: A Retrospective Cohort Study. Clinical Interventions in Aging. 2024;19:1919-1938. DOI: 10.2147/CIA.S487629.

9. Wang X, Tao J, Zhong Y, Yao Y, Wang T, Gao Q, Xu G, Lv T, Li X, **Sun D**, Cheng Z, Liu M, Xu J, Wu C, Wang Y, Wang R, Zheng B, Yan M. Nadir Hemoglobin Concentration After Spinal Tumor Surgery: Association With Risk of Composite Adverse Events. Global Spine Journal. Published online November 2, 2023; print: 2025;15(2):800-807. DOI: 10.1177/21925682231212860.

10. Wang T, Gao Q, Yao Y, Luo G, Lv T, Xu G, Liu M, Xu J, Li X, **Sun D**, Cheng Z, Wang Y, Wu C, Wang R, Zou J, Yan M. Causal relationship between obesity and iron deficiency anemia: a two-sample Mendelian randomization study. Frontiers in Public Health. 2023;11:1188246. DOI: 10.3389/fpubh.2023.1188246.

11. Ming Y, Zhang F, Yao Y, Cheng Z, Yu L, **Sun D**, Sun K, Yu Y, Liu M, Ma L, HuangYang Y, Yan M. Large volume acute normovolemic hemodilution in patients undergoing cardiac surgery with intermediate-high risk of transfusion: A randomized controlled trial. Journal of Clinical Anesthesia. 2023;87:111082. DOI: 10.1016/j.jclinane.2023.111082.

12. Li X, Wang R, **Sun D**, Yao Y, Wang T, Luo G, Liu M, Xu J, Cheng Z, Gao Q, Wang Y, Wu C, Xu G, Lv T, Zou J, Yan M. Risk Factors for Hypocoagulability After Cardiac Surgery: A Retrospective Study. Clinical and Applied Thrombosis/Hemostasis. 2023;29:10760296231209927. DOI: 10.1177/10760296231209927.

13. **Sun D**, Lu F, Sheldon A, Jiang X, Ferriero DM. Neuronal deficiency of hypoxia-inducible factor 2α increases hypoxic-ischemic brain injury in neonatal mice. Journal of Neuroscience Research. 2021;99(11):2964-2975. **(First author)** DOI: 10.1002/jnr.24943.

14. Shao R, **Sun D**, Hu Y, Cui D. White matter injury in the neonatal hypoxic-ischemic brain and potential therapies targeting microglia. Journal of Neuroscience Research. 2021;99(4):991-1008. DOI: 10.1002/jnr.24761.

15. Li Y, **Sun D**, Cui D. Research progress in the relationship between endosomal sorting complexes required for transport and autophagy. Journal of Shanghai Jiao Tong University (Medical Science). 2021;41(9):1256-1260. DOI: 10.3969/j.issn.1674-8115.2021.09.019.

16. Li Y, Hu Y, **Sun D**, Cui D. Effects of different rewarming rates on neuron autophagy in the cardiac arrest/resuscitation rat model treated with hypothermia. Journal of Shanghai Jiao Tong University (Medical Science). 2020;40(11):1454-1460. DOI: 10.3969/j.issn.1674-8115.2020.11.003.

17. Hu Y, **Sun D**, Li Y, Wang X, Jiang W, Shi H, Cui D. Increased PINK1/Parkin-mediated mitophagy explains the improved brain protective effects of slow rewarming following hypothermia after cardiac arrest in rats. Experimental Neurology. 2020;330:113326. **(Co-first author)** DOI: 10.1016/j.expneurol.2020.113326.

18. Hu Y, Guo Y, Wang X, Li Y, **Sun D**, Cui D. Effects of the Incidence Density of Fever (IDF) on Patients Resuscitated From In-Hospital Cardiac Arrest: A Mediation Analysis. Frontiers in Medicine. 2020;7:86. DOI: 10.3389/fmed.2020.00086.

19. Wang X, **Sun D**, Hu Y, Xu X, Jiang W, Shang H, Cui D. The roles of oxidative stress and Beclin-1 in the autophagosome clearance impairment triggered by cardiac arrest. Free Radical Biology and Medicine. 2019;136:87-95. **(Co-first author)** DOI: 10.1016/j.freeradbiomed.2018.12.039.

20. Xu X, Zhu Q, Niu F, Zhang R, Wang Y, Wang W, **Sun D**, Wang X, Wang A. A2BAR activation attenuates acute lung injury by inhibiting alveolar epithelial cell apoptosis both in vivo and in vitro. American Journal of Physiology-Cell Physiology. 2018;315(4):C558-C570. DOI: 10.1152/ajpcell.00294.2017.

21. **Sun D**, Wang W, Wang X, Wang Y, Xu X, Ping F, Du Y, Jiang W, Cui D. bFGF plays a neuroprotective role by suppressing excessive autophagy and apoptosis after transient global cerebral ischemia in rats. Cell Death & Disease. 2018;9(2):172. **(First author)** DOI: 10.1038/s41419-017-0229-7.

22. Xu X, Zhu Q, Zhang R, Wang Y, Niu F, Wang W, **Sun D**, Wang A. ITRAQ-Based Proteomics Analysis of Acute Lung Injury Induced by Oleic Acid in Mice. Cellular Physiology and Biochemistry. 2017;44(5):1949-1964. DOI: 10.1159/000485885.

23. Wang X, **Sun D**, Cui D. Critical care progress of traumatic spinal cord injury. Chin J TCM WM Crit Care. (2017) 24: 552-555.

24. Cui D, **Sun D**, Wang X, Yi L, Kulikowicz E, Reyes M, Zhu J, Yang ZJ, Jiang W, Koehler RC. Impaired autophagosome clearance contributes to neuronal death in a piglet model of neonatal hypoxic-ischemic encephalopathy. Cell Death & Disease. 2017;8(7):e2919. DOI: 10.1038/cddis.2017.318.

SELECTED SCIENTIFIC PRESENTATIONS

2024. Sun D, Yao Y, Yan M. Arterial-venous oxygen difference as a potential indicator for erythrocyte transfusion during cardiac surgery: a prospective observational study. Selected for oral presentation · American Society of Anesthesiologists Annual Meeting · Philadelphia, PA.

2019. Sun D, Wang X, Hu Y, Cui D. Angiopep-2 modified PEGylated 2-methoxyestradiol micelles contributes to neuroprotection in a murine cardiac arrest model. Oral presentation · Best of Abstracts: Basic Science (one of 10 selected abstracts from more than 1,000 submissions) · American Society of Anesthesiologists Annual Meeting · Orlando, FL.

PROFESSIONAL SERVICE AND LEADERSHIP

Volunteer Service - Ad Hoc Peer Reviewer, 2024 - Present
Completed 53 peer-review assignments across 17 journals, as of September 2026, including International Journal of Surgery, Anesthesiology and Perioperative Science, Journal of Clinical Anesthesia, Drug Design, Development and Therapy, International Journal of Molecular Sciences, Journal of Clinical Medicine, Brazilian Journal of Anesthesiology, Abdominal Radiology, Anesthesia Research, Biomedicines, Emergency Care and Medicine, Healthcare, Journal of Anesthesia and Translational Medicine, Journal of Personalized Medicine, Journal of Vascular Diseases, Methods and Protocols, and Open Medicine.

President, Graduate Student Union, Shanghai Jiao Tong University Affiliated Sixth Hospital, China, 2017 - 2018.

Vice President, Student Union, Wenzhou Medical University, China, 2014 - 2015.

Public CV updated October 14, 2026. Based on the September 2026 CV; observerships and clinic contributions updated from information supplied by Dawei Sun.